Quiz II

Name: **Entry Number:**

1. (5 marks) You are given a skip list consisting of n elements where the probability that an element gets promoted to one level up is p (instead of $1/2$ in the usual skip list). What is the expected time to search for an element in the skip list? Give a short explanation. Do not hide terms depending on p in the $O()$ notation, i.e., do not assume that p is a constant.

2. (5 marks) Recall that a family $\mathcal{H}$ of hash functions mapping a universe U to $\{0, 1, \ldots, m-1\}$ is called *universal* if for every distinct $x, y \in U$,

$$\Pr_{h \in \mathcal{H}}[\, h(x) = h(y)\,] \leq \frac{1}{m}.$$

   Let $U = \{0, 1, 2, 3\}$ and let

$$\mathcal{H} = \{h_1(x) = x \bmod 2,\ h_2(x) = (x+1) \bmod 2\}.$$

   Note that $\mathcal{H}$ consists of two hash functions h_1 and h_2 as defined above. Each h_i maps U into $\{0, 1\}$. Is $\mathcal{H}$ a universal family of hash functions? Justify your answer.

3. (5 marks) Let A be an $m \times n$ matrix and let A have rank n. Let $v_1, \ldots, v_k \in \Re^n$ be linearly independent vectors. Are the vectors $Av_1, \ldots, Av_k$ linearly independent? Give reasons.

4. (5 marks) Let V be the vector space $\Re^4$. Let W_1 be the subspace spanned by $\{(1,0,1,0),(0,1,0,1)\}$ and W_2 be the subspace spanned by $\{(1,1,0,0),(0,0,1,1)\}$. What are the dimensions of $W_1 \cap W_2$ and $W_1 + W_2$? Give reasons.

5. (5 marks) Let $V = \Re^2$ with bases

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}, \quad \mathcal{C} = \left\{ \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 3 \end{bmatrix} \right\}.$$

Suppose a vector v has coordinates $(4, 5)$ in basis $\mathcal{B}$. What are its coordinates in basis $\mathcal{C}$. Give reasons.